

TODP V1 Architecture Design Document (ADD)

Sections 11–15

Status: FROZEN

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11. Discovery Architecture

Purpose

Discover opportunities, events, scholarships, government programs, courses, tools, resources, institutions and services using evidence-based retrieval.

Discovery Workflow

User Query

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Context Identification

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Metadata Filtering

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Authority Discovery

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Hybrid Retrieval

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Evidence Extraction

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Source Validation



Citation Validation



Candidate Results



Ranking



Audit



Display

Discovery Scope Expansion

Discovery must occur in the following order:

User Context



Institution



Local



District



State

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National

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International

Discovery Sources

Examples:

- Government Websites
 - University Websites
 - NGO Websites
 - Official Organization Websites
 - Official Program Pages
 - Official Event Pages
 - Official Documentation
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Discovery Outputs

Every discovery result must contain:

- Source ID
- Citation ID
- Retrieval Timestamp
- Evidence Chain ID
- Version

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12. Ranking Architecture

Purpose

Provide deterministic, reproducible ranking.

No hidden ranking logic.

Ranking Formula

Total Cost Score

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Money Cost

-

Time Cost

-

Distance Cost

Money Cost

Includes:

- Program Fee
 - Registration Fee
 - Tool Cost
 - Service Cost
 - Subscription Cost
-

Time Cost

Includes:

- Travel Time
 - Waiting Time
 - Completion Time
 - Access Time
-

Distance Cost

Includes:

- Distance From User Context
-

Online Opportunity Rule

For online opportunities:

Distance Cost = 0

Travel Time = 0

Ranking Workflow

Candidate Results



Cost Extraction



Money Cost Calculation



Time Cost Calculation



Distance Cost Calculation



Ranking Score



Audit



Display

Ranking Audit Requirements

Verify:

- Ranking Reproducibility
- Ranking Consistency
- Ranking Traceability

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13. Registration Architecture

Purpose

Assist users in registration without taking actions on behalf of users.

Supported Functions

Registration Discovery

Retrieve:

- Registration URL
 - Registration Deadline
 - Registration Requirements
-

Registration Assistance

Allowed:

- Autofill
- Form Preparation
- User Guidance

Not Allowed:

- Auto Submission
 - Automated Consent
 - Automated Payment
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Registration Workflow

Recommendation

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Evidence Review

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User Approval

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Autofill

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User Review

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User Submission

Registration Metadata

Store:

- Registration ID
- Opportunity ID
- Timestamp
- Status
- Audit Record

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14. Calendar Architecture

Purpose

Help users schedule opportunities.

Calendar usage requires explicit user consent.

Supported Calendars

Examples:

- Google Calendar
- Microsoft Outlook
- Other Supported Calendars

Calendar Functions

Availability Check

Determine whether an event conflicts with existing calendar entries.

Scheduling Assistance

Suggest:

- Available Time Slots
- Event Placement
- Conflict Warnings

Calendar Entry Creation

Allowed only after:

- User Consent
- User Confirmation

Calendar Workflow

Opportunity Selected



Availability Check



Conflict Detection

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Scheduling Recommendation

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User Approval

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Calendar Entry Creation

Calendar Metadata

Store:

- Calendar Entry ID
- Event ID
- Timestamp
- Sync Status

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15. Security, Consent and Governance Architecture

Purpose

Protect users, preserve auditability and maintain system integrity.

Consent Principles

User consent is required for:

- Calendar Access
 - Registration Autofill
 - Location Usage
 - Account Linking
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Governance Principles

The system must:

- Preserve Provenance
 - Preserve Audit Records
 - Preserve Evidence Chains
 - Preserve Recommendation History
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Recommendation Governance

Every recommendation must be:

- Traceable
 - Auditable
 - Reproducible
 - Explainable
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Evidence Governance

Every recommendation must contain:

- Source
 - Citation
 - Timestamp
 - Evidence Chain
-

Hallucination Governance

The system must continuously verify:

- Missing Sources
 - Missing Citations
 - Unsupported Claims
 - Unsupported Recommendations
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Governance Rule

No Evidence

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No Strong Recommendation

Audit Questions

The platform must continuously answer:

Can every recommendation be traced?

Can every ranking be reproduced?

Can every source be audited?

Can every citation be verified?

Can every hallucination be detected?

Can every retrieval be reproduced?

V1 Governance Motto

Trace Everything.

Audit Continuously.

Trust Evidence.

Preserve Provenance.

Detect Hallucinations.

Never Recommend Without Evidence.

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END OF ADD V1 SECTIONS 11-15